Separability and decomposition in mechanism design with transfers *

Debasis Mishra, Swaprava Nath, and Souvik Roy †

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Abstract

In private values quasi-linear environment, we consider problems where allocation decisions along multiple components have to be made. Every agent has additively separable valuation over the components. We show that every neutral and dominant strategy implementable allocation rule in this problem is a *neutral component affine maximizer*, which assigns non-negative weight vectors to agents in each component and chooses an alternative in each component by maximizing the weighted sum of valuations in that component. A corollary of our result is that every neutral and dominant strategy implementable allocation rule can be *almost* decomposed (modulo tie-breaking) into dominant strategy implementable allocation rules along each component.

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†Debasis Mishra, Indian Statistical Institute, New Delhi, Email: dmishra@isid.ac.in; Swaprava Nath, Carnegie Mellon University, Email: swapravn@cs.cmu.edu; Souvik Roy, Indian Statistical Institute, Kolkata, Email: souvik_roy2004@yahoo.com

1 Introduction

Suppose a planner needs to choose various types of public goods for a city: a stadium location among a set of locations, a school with a particular medium of teaching among a set of teaching mediums, a mode of public transport among various modes of transport etc. Transfers (in the form of taxes or subsidies) are allowed and agents have (i) additively separable value over different types of public good and (ii) quasi-linear utility over transfers. How should the planner choose different types of public good?

We investigate the class of dominant strategy incentive compatible mechanisms in such environments - our focus is exclusively on deterministic mechanisms. These kind of settings are typically characterized by alternatives which have multiple dimensions or components - types of public good in the above example. Agents have additively separable valuations, which are defined by a valuation function over alternatives in each component, and then, the value for an alternative is just the sum of valuations of individual components. For instance, in the example, value for an alternative which consists of (a^1) a stadium in downtown, (a^2) a French medium school, and (a^3) a monorail transport system is calculated as the sum of values of a^1, a^2, and a^3.

In a strategic voting environment, where transfers are precluded, such separability usually implies decomposition of the mechanism, i.e., we can employ a separate mechanism on each component that only considers preference information on that component and then the decisions on various components can be aggregated (Barberà et al., 1991; Barbera et al., 1993; Le Breton and Sen, 1999; Le Breton and Weymark, 1999; Weymark, 1999; Svensson and Torstensson, 2008). [1]

We show that such decomposition is not possible in our setting with transfers. In particular, in quasi-linear environment with additively separable valuations over components, a dominant strategy implementable allocation rule is not necessarily decomposable. [2] This is not surprising because introduction of transfers usually expands the set of incentive compatible mechanisms, and in the process, we get new mechanisms which are not decomposable. Surprisingly, our main result shows that the decomposability result can be restored if we assume *neutrality* of the allocation rule. In particular, we show that every dominant strategy implementable and neutral allocation rule must be a *neutral component affine maximizer (NCAM)*. Conversely, every NCAM is implementable under a mild tie-breaking condition.

An NCAM allocation rule specifies for each component a vector of non-negative weights

[1] Usually, these results require additional mild conditions such as *unanimity* or *ontoness*.

[2] An allocation rule is dominant strategy implementable if there exists a payment function such that the corresponding mechanism is dominant strategy incentive compatible.

(not all zero) for the agents. Then, for each component, it computes weighted sum of valuations of agents and chooses an alternative that maximizes this weighted sum. Because of tie-breaking issues, this does not give us complete decomposability of the allocation rule but implies decomposability in a generic sense.

Decomposability is a desirable property in this environment. In our example, the planner would ideally delegate the decision of choosing a school to experts in education who will take their decision independently. In particular, it makes more sense that they will only consider values of citizens for various medium of schools and not worry about how they value locations of a stadium (most likely, they will struggle to evaluate values of agents on any dimension except schooling choice). Our result establishes that even in quasi-linear environment with transfers, the desirable decomposition property is true if we assume neutrality.

Neutrality is a sufficient condition for our main result - we give an example to show that there are non-neutral and implementable allocation rules that cannot be decomposed. We feel that neutrality is a very natural condition in the examples we consider. It requires that the social planner has no bias over alternatives in each component. For instance, the social planner does not care whether an English medium or a French medium or a Hindi medium school is opened in a city - it will take the decision solely based on the valuations of the citizens. In that sense, neutrality reflects an unbiased planner who is only a representative of the agents.

We also show that for every NCAM allocation rule, there is a decomposed payment rule (i.e., total payment is the sum of the payments of each component) that implements such NCAM allocation rule. Standard revenue equivalence result then pins down the entire class of payment rules that can implement NCAMs - these are payment rules which will differ from the decomposed payment rule by an additive constant, which does not depend on agent's own valuation (Chung and Olszewski, 2007). However, this implies that there may be other non-decomposable payment rules that may implement an NCAM allocation rule. Hence, we cannot be assured of decomposition of every dominant strategy incentive compatible *mechanism* even though the allocation rule can be decomposed.

Our result contributes to two strands of literature. As discussed earlier, it complements the literature on decomposition of mechanisms in separable environments without transfers - for instance, see Barberà et al. (1991) for the case of two alternatives in each component, Barbera et al. (1993) for multidimensional single-peaked preferences, Le Breton and Sen (1999) for a general result covering many preference domains, Le Breton and Weymark (1999) for a model with a continuum of alternatives, Svensson and Torstensson (2008) for a component-wise dictatorship result when preferences in each component is unrestricted,

and Weymark (1999) for a for a detailed survey with a general result. Probably, the closest model in this literature is the one of Svensson and Torstensson (2008), who consider the public good provision problem but do *not* allow for transfers. As a result, every incentive compatible and onto mechanism in their model is a *component-wise dictatorship*. Our result can be thought as a generalization of their result in the multiple public good provision model with transfers but with quasilinear utility. Indeed, transfers are a natural tool for a planner in this environment and our result clarifies the exact nature of possibility and decomposition in the presence of transfers.

Next, it contributes to a new literature in mechanism design about extending a seminal result of Roberts (1979). In an important paper, Roberts (1979) showed that every onto and dominant strategy implementable allocation rule in an unrestricted type space in quasi-linear utility environment must be an *affine maximizer*. [3] Several authors establish simpler proofs (Lavi et al., 2009; Dobzinski and Nisan, 2009; Vohra, 2011) or extend his result to other restricted type space - for instance, Mishra and Sen (2012) show that Roberts theorem holds in certain bounded type spaces if neutrality is assumed; Carbajal et al. (2013) extend his result to infinite set of alternatives; Jehiel et al. (2008) extend it to certain interdependent valuation environment; Nath and Sen (2015) extend Roberts' result to a specific private good allocation problem assuming an additional condition called *non-bossiness*. All these papers find new type spaces where implementability along with other reasonable conditions imply affine maximization. Mishra and Quadir (2014) provide an analogue of Roberts' theorem in the private values single object auction environment. Their result captures a larger class of allocation rules than just affine maximizers. Similarly, Dobzinski and Nisan (2011) show the existence of non-affine maximizers in multi-unit auction environment.

The richness of type space required in these papers is absent in the type space we consider. An important aspect of our type space (due to additive separability) is that altering the valuation of one alternative involves modifying one of the components, and this in turn changes the valuations of many alternatives. As a result, we lose the freedom of modifying types along one of the alternatives without disturbing the values of other alternatives - this is a crucial element of getting to Roberts' result, which is exploited in the literature. As a consequence of absence of this richness, we get our *new* class of implementable allocation rules - NCAM allocation rules. Further, none of the existing results imply our result and the set of NCAM allocation rules captures a larger class of allocation rules than the neutral affine maximizers in Roberts (1979) - we discuss this issue in detail later.

[3] This result requires at least three alternatives. If there are two alternatives, there are implementable allocation rules that are not affine maximizers. Marchant and Mishra (2015) provide a complete characterization for the two-alternatives case.

The rest of the paper is organized as follows. We introduce our separable type space in Section 2 along with a motivating example in Section 2.1. The main result is presented in Section 3 followed by some remarks in Section 3.1. The proof of the main result is given in Section 4. We conclude the paper in Section 5 and provide a missing proof in the Appendix at the end.

2 The Separable Type Space

We begin by formally defining our model of separable type space with transfers. Let $N = \{1, \ldots, n\}$ be the set of agents. Agents need to decide on various types of public goods. A set $J = \{1, \ldots, m\}$ reflects the types of public good. We will also call the elements of J the *components*. For each component $j \in J$, let A^j be the (finite) set of available public goods of type j. We assume that for all $j \in J$, we have $|A^j| \geq 3$. A public good along each component must be chosen. Hence, the set of alternatives is denoted by

$$A := A^1 \times \ldots \times A^m.$$

An alternative $a \in A$ will be equivalently denoted as $(a^1, \ldots, a^m)$, where for every j, $a^j \in A^j$ is the public good of type j in a.

The *type* of an agent $i \in N$ is a valuation function $v_i : A \to \mathbb{R}$. We assume that the valuations of agents are separable in the following sense.

Definition 1 *A valuation function v_i of agent i is* **(additively) separable** *if there exists a collection of vectors $(v_i^1, \ldots, v_i^m)$, where $v_i^j \in \mathbb{R}^{|A^j|}$ for each $j \in J$, such that for every $a \in A$,*

$$v_i(a) = \sum_{j \in J} v_i^j(a^j).$$

Here, v_i^j will be referred to as the valuation of agent i for the alternatives in the j-th component.

A profile of valuation functions $(v_1, \ldots, v_n)$ will be denoted by $\mathbf{v}$. Further, for every $i \in N$, we will sometimes denote its valuation v_i as $(v_i^1, \ldots, v_i^m)$.

The **type space** of each agent can be restricted by restricting the values of the valuation functions along each component. In general, we assume that for every $i \in N$ and for every $j \in J$, the valuations of alternatives in the j-component lie in $\mathbb{R}_{++}$. [4] Let $V_i^j \equiv \mathbb{R}_{++}^{|A^j|}$ be the

[4] We denote the set of non-negative and positive real numbers by $\mathbb{R}_+$ and $\mathbb{R}_{++}$ respectively.

set of all valuation vectors of agent i for the j-th component such that for each $v_i^j \in V_i^j$, we have $v_i^j(a^j) \in \mathbb{R}_{++}$ for all $a^j \in A^j$. Hence, the type space of each agent $i \in N$ is

$$V_i \equiv V_i^1 \times \ldots \times V_i^m,$$

where $V_i^j \equiv \mathbb{R}_{++}^{|A^j|}$ for each $j \in J$. We refer to this type space assumption as **separable type space**.

Sometimes, we will look at the valuations of all the agents $\mathbf{v}$ restricted to a particular component j. In that case, we will refer this restricted profile along the j-th component as v^j. The separable type space automatically induces a set of such restricted valuation profiles along the j-th component, and we will denote this as V^j.

We allow transfers in our model but assume that preference over transfer is quasi-linear. Suppose alternative $a \equiv (a^1, \ldots, a^m)$ is chosen and agent i is paid an amount p_i, then the net utility of agent i with valuation v_i is

$$\sum_{j \in J} v_i^j(a^j) + p_i.$$

A **allocation rule** is a map $F : V_1 \times \ldots \times V_n \to A$. At any valuation profile $\mathbf{v}$, the public good of type j chosen by the allocation rule F is denoted by $F^j(\mathbf{v})$.

A **payment function** of agent i is a map $p_i : V_1 \times \ldots \times V_n \to \mathbb{R}$. Note that payment of agent i need *not* be separable across the public good types.

A mechanism is an allocation rule and a collection of payment functions: $(F, p_1, \ldots, p_n)$.

Definition **2** *An allocation rule F is* **implementable** *(in dominant strategies) if there exists $(p_1, \ldots, p_n)$ such that for every $i \in N$, for every $v_{-i} \in \times_{k \neq i} V_k$, for every $v_i, v_i' \in V_i$, we have*

$$\sum_{j \in J} v_i^j(F^j(v_i, v_{-i})) + p_i(v_i, v_{-i}) \geq \sum_{j \in J} v_i^j(F^j(v_i', v_{-i})) + p_i(v_i', v_{-i}).$$

In this case, we say $(F, p_1, \ldots, p_n)$ is a **dominant strategy incentive compatible (DSIC)** *mechanism.*

From a planner's perspective, for each component $j \in J$, it can have a separate allocation rule $f^j : V^j \to A^j$ to decide on the alternative of that component. It chooses alternatives in each component using an implementable *marginal allocation rule* on that component. It can then aggregate the decisions on each component. Hence, decomposability of an allocation rule would be equivalent to existence of such *marginal* allocation rules over each component.

Definition 3 *An allocation rule F is* **decomposable** *if there exists a collection of implementable marginal allocation rules $(f^1, \ldots, f^m)$, where $f^j : V^j \to A^j$ for every $j \in J$, such that for all $\mathbf{v} \in V_1 \times \ldots \times V_n$, we have*

$$F^j(\mathbf{v}) = f^j(v_1^j, \ldots, v_n^j) \quad \forall\ j \in J.$$

Given the additive separability of valuations, it is natural to suspect that an implementable allocation rule may be decomposable. This is the main question we aim to answer:

Is every implementable allocation rule decomposable?

2.1 A Motivating Example

We start with an example which shows that implementability need not imply decomposability, and this in turn motivates us to ask the same question under an additional condition. Our example has two agents and two components. Component 1 has alternatives $\{a^1, b^1, c^1\} = A^1$ and component 2 has alternatives $\{a^2, b^2, c^2\} = A^2$. Hence, $A = A^1 \times A^2$. We will define an implementable allocation rule in this problem which is not decomposable.

Define a map $\kappa : A \to \mathbb{R}$ as follows: $\kappa(a^1, a^2) = 0.95, \kappa(b^1, b^2) = 1$ and $\kappa(x) = 0$ for all alternatives $x \notin \{(a^1, a^2), (b^1, b^2)\}$. Now, at every valuation profile $\mathbf{v} \equiv (v_1, v_2)$,

$$F(\mathbf{v}) \in \arg\max_{x \in A} \big[v_1(x) + v_2(x) + \kappa(x)\big],$$

where for every $i \in N$, $v_i(x) = v_i^1(x^1) + v_i^2(x^2)$. We break ties by using a strict ordering over the set of alternatives - in particular, in case of multiple alternatives maximizing, we pick the highest ranked alternative according to this tie-breaking ordering which belongs to the set of maximizers. Such allocation rules are implementable using payment rules which are general versions of Groves payments (Carbajal et al., 2013).

We argue that such an allocation rule is not decomposable. Indeed, if F was decomposable, then the change in valuations in one component would not change the outcome in the other component. We show that this property is violated. Consider the following pair of valuation profiles which differs in the valuation in the second component. Denote the first valuation profile as $\mathbf{v} \equiv (v_1, v_2)$ and the second one as $\bar{\mathbf{v}} \equiv (\bar{v}_1, \bar{v}_2)$. In particular, all valuations in both the profiles are arbitrarily close to zero except agent 1's valuation in $\bar{v}_1$ on alternative a^2 of component 2, which is set at

$$\bar{v}_1^2(a^2) = 0.1.$$

By construction,

$$F(\mathbf{v}) = (b^1, b^2) \text{ and } F(\bar{\mathbf{v}}) = (a^1, a^2).$$

Note that $\mathbf{v}$ and $\bar{\mathbf{v}}$ only differ in the valuation in the second component, but the outcome of F in the first component changed from $\mathbf{v}$ to $\bar{\mathbf{v}}$. This shows that F is implementable but not decomposable.

Our main result below shows that in the class of additively separable valuations, if the allocation rule is *neutral* with respect to the set of alternatives in each component, then implementability implies decomposability in a generic sense (i.e, except for some tie-breaking issues).

3 The Main Result

We require the following neutrality property. For every $j \in J$, let π^j be a permutation of A^j. Given a profile of valuations $\mathbf{v}$, we can permute it along j by applying the permutation π^j - note, we only permute the component j of all the agents. With a small abuse of notation, we also denote the permutation of the valuation profile induced by the permutation of the alternative with the same symbol, i.e., $\pi^j(\mathbf{v})$ denotes a valuation profile where the valuations at $\pi^j(a^j) \in A^j$ are the same as the valuations in $\mathbf{v}$ at a^j. For example, suppose that there are two agents and two components and the alternative sets are $A^1 = \{a^1, b^1, c^1\}, A^2 = \{a^2, b^2, c^2\}$. If $\pi^1(a^1) = b^1, \pi^1(b^1) = a^1, \pi^1(c^1) = c^1$, then an example of $\mathbf{v}$ and $\pi^1(\mathbf{v})$ will be:

	A^1			A^2		
$\mathbf{v}$	a^1	b^1	c^1	a^2	b^2	c^2
v_1	2	3	4	7	6	5
v_2	5	1	2	8	4	3

	A^1			A^2		
$\pi^1(\mathbf{v})$	a^1	b^1	c^1	a^2	b^2	c^2
v_1	3	2	4	7	6	5
v_2	1	5	2	8	4	3

Note that the separable type space assumption implies that if $\mathbf{v}$ is in the type profile space, then $\pi^j(\mathbf{v})$ is also in the type profile space.

Definition 4 *An allocation rule F is* **neutral** *if for all $j \in J$, for all permutations π^j, and for all $\mathbf{v}$ with $\pi^j(\mathbf{v}) \neq \mathbf{v}$, we have*

$$\pi^j(F^j(\mathbf{v})) = F^j(\pi^j(\mathbf{v})).$$

Neutrality says that if the names of the alternatives are changed along a component, then the outcome along that component must be changed accordingly. In the earlier example,

this implies that if $F^1(\mathbf{v}) = a^1$, then $F^1(\pi^1(\mathbf{v})) = b^1$. [5]

We are now ready to state our main result, whose proof is postponed to Section 4.

THEOREM 1 **(Neutral Component Affine Maximization)** *Suppose $F : V_1 \times \ldots \times V_n \to A$ is a neutral allocation rule, where V_i is a separable type space for each $i \in N$. If F is implementable, for every $j \in J$, there exists weight vectors $\lambda^j \in \mathbb{R}^n_+ \setminus \{\mathbf{0}\}$ such that for all $\mathbf{v}$, we have*

$$F^j(\mathbf{v}) \in \arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i^j v_i^j(a^j).$$

Theorem 1 explicitly points out the parameters that can be used in a neutral and implementable allocation rule. We denote these allocation rules as **neutral component affine maximizers (NCAM)**. An NCAM allocation rule assigns weights on each component and decides on the alternative of a component using the corresponding weights.

It is easy to show that if ties are not handled properly, then an NCAM allocation rule may no longer be implementable. [6] We introduce a mild tie-breaking condition that ensures implementability of NCAM allocation rules.

DEFINITION 5 *An NCAM allocation rule F with weights $\{\lambda^j\}_{j \in J}$ satisfies* **independence of irrelevant agents (IIA)** *if for every $j \in J$ and for every $i \in N$ with $\lambda_i^j = 0$, we have $F^j(\mathbf{v}) = F^j(\mathbf{v}')$ if $\mathbf{v}$ and $\mathbf{v}'$ differ only in the valuation of agent i.*

Now, we define a payment function that implements an NCAM allocation rule satisfying IIA. First, we define a payment function for each component j as follows: for every $i \in N$ and for every $\mathbf{v}$

$$p_i^j(\mathbf{v}) = \begin{cases} \frac{1}{\lambda_i^j} \sum_{k \neq i} \lambda_k^j v_k^j(F^j(\mathbf{v})) & \text{if } \lambda_i^j > 0 \\ 0 & \text{if } \lambda_i^j = 0 \end{cases}$$

Now, we can define a payment function for agent i as: for every $\mathbf{v}$,

$$p_i(\mathbf{v}) = \sum_{j \in J} p_i^j(\mathbf{v}).$$

The lemma below shows that this allocation rule implements an NCAM satisfying IIA. The proof follows standard arguments and is given in the Appendix at the end.

[5] A reader may relate this definition of neutrality to a more commonly used definition, where the permutation $\pi : A \to A$ is defined over all alternatives and not restricted to any components. Since any such permutation π can be constructed by a sequence of component-wise permutations, our definition of neutrality is *equivalent* to this definition.

[6] We skip a formal example to illustrate this fact. The issue is similar to the non-implementability of affine maximizers of Roberts (1979) if ties are not broken carefully - see detailed discussions in Carbajal et al. (2013).

Lemma 1 *If F is an NCAM allocation rule satisfying IIA, then $(p_1, \ldots, p_n)$ implements F.*

Unfortunately, an NCAM allocation rule is not completely decomposable. The issue is the existence of multiple maximizers at certain valuation profiles, i.e., how we handle tie-breaking in the maximization above. Since Theorem 1 cannot pin down how to select an alternative for a component if there are multiple maximizers, it leaves room for the failure of decomposability. We give an example to illustrate this.

Consider two agents and two components. The alternatives in component 1 are a^1, b^1, c^1. Consider a valuation profile whose component 1 has the following valuations:

$$v_1^1(a^1) = 2, v_1^1(b^1) = 0, v_1^1(c^1) = 1,$$

$$v_2^1(a^1) = 4, v_2^1(b^1) = 5, v_2^1(c^1) = 1.$$

The NCAM weights for component 1 are $\lambda_1^1 = 1, \lambda_2^1 = 2$. Both a^1 and b^1 maximize the weighted sum in the NCAM. By Lemma 1, we can pick either a^1 or b^1 on this component, and the NCAM will still be implementable (note that the NCAM trivially satisfies IIA in this case). In particular, we can choose a^1 on component 1 when valuations of other components is v^2 but b^1 when valuations of other components is $\bar{v}^2 \neq v^2$. This will violate our definition of decomposability. However, we argue that such failure of decomposability happens only in these kind of valuation profiles which has ties in the NCAM maximization, and forms a set of measure zero over the set of all valuation profiles.

This example motivates us to consider a slightly weaker version of decomposability.

Definition 6 *An allocation rule F is* **decomposable almost everywhere** *if there exists a collection of implementable marginal allocation rules $(f^1, \ldots, f^m)$, where $f^j : V^j \to A^j$ for every $j \in J$, such that for every $j \in J$ there exists $X^j \subseteq V^j$ with $V^j \setminus X^j$ having zero measure and for all v^{-j}, we have*

$$F^j(v^j, v^{-j}) = f^j(v^j) \quad \forall\ v^j \in X^j.$$

A simple corollary of Theorem 1 is the following almost decomposability result.

Corollary 1 **(Almost Decomposition)** *Suppose $F : V_1 \times \ldots \times V_n \to A$ is a neutral allocation rule, where V_i is a separable type space for each $i \in N$. Then, if F is implementable, it is decomposable almost everywhere.*

Proof: By Theorem 1, for every $j \in J$, there exists weight vectors $\lambda^j \in \mathbb{R}_+^n \setminus \{\mathbf{0}\}$ such that for all $\mathbf{v}$, we have

$$F^j(\mathbf{v}) \in \arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i^j v_i^j(a^j).$$

Fix a component $j \in J$. Let $X^j \subseteq V^j$ be defined as follows:

$$X^j := \{v^j \in V^j : |\arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i^j v_i^j(a^j)| = 1\}.$$

Note that $V^j \setminus X^j$ has zero measure. Now, fix any $\bar{v}^{-j}$, and consider the restriction of F to the j-th component by fixing $\bar{v}^{-j}$. In particular, define $f^j : V^j \to A^j$ as follows. For every $v^j \in V^j$, let

$$f^j(v^j) := F^j(v^j, \bar{v}^{-j}).$$

By construction, for every $v^j \in X^j$ and any v^{-j}, v'^{-j}, we have

$$F^j(v^j, v^{-j}) = F^j(v^j, v'^{-j}).$$

Hence, for all $v^j \in X^j$ and all v^{-j}, we have

$$f^j(v^j) = F^j(v^j, v^{-j}),$$

establishing the claim that F is decomposable almost everywhere. ■

3.1 Remarks

Roberts' Theorem. Roberts (1979) worked in an environment where each alternative cannot be separated into components (or alternatively, there is only one component for every alternative). He showed that in such an environment if $|A| \geq 3$ and type space of each agent is $\mathbb{R}^{|A|}$ (i.e., the set of all possible type vectors), then every onto and implementable allocation rule is an *affine maximizer*, i.e., there exists weight vectors $w \in \mathbb{R}^n_+ \setminus \{0\}$ and a map $\kappa : A \to \mathbb{R}$ such that at every valuation profile $\mathbf{v}$, we have [7]

$$F(\mathbf{v}) \in \arg\max_{a \in A} \big[\sum_{i \in N} w_i v_i(a) + \kappa(a)\big]. \tag{1}$$

Under a mild tie-breaking condition, similar to the IIA condition in Definition 5, every affine maximizer is also implementable (Mishra and Sen, 2012) - see Carbajal et al. (2013) for a complete characterization.

Roberts theorem does not apply to our setting because our alternatives have components and valuation of an alternative can be additively separated across values over components. As a result, for every alternative a, changing the value of alternative a also changes the

[7] Since this is a setting where there are no components, valuation vector of each agent i is just $v_i \in \mathbb{R}^{|A|}$ with $v_i(a)$ denoting the value for any alternative a.

values of all alternatives which uses some of the same component alternatives. For instance, consider an example with two components and suppose we change the value of alternative $a \equiv (a^1, a^2)$. Then, this will also change the values of alternative, say, (a^1, b^2) and (b^1, a^2). This destroys the richness required to get the affine maximizer result in Roberts (1979).

Of course, the affine maximizers (with appropriate tie-breaking) are implementable in our model also. The NCAM allocation rules that we identify are a new class of allocation rules - they are very specific to our environment with alternatives having components and agents having additively separable valuations over components. Call affine maximizers whose κ terms are all zero in Equation 1 as *neutral* affine maximizers. It is easy to see that in our model, a neutral affine maximizer is also an NCAM - this is because for all i, $w_i v_i(a) = \sum_{j \in J} w_i v_i^j(a^j)$ and hence, the affine maximization

$$\sum_{i \in N} \sum_{j \in J} w_i v_i^j(a^j) = \sum_{j \in J} \sum_{i \in N} w_i v_i^j(a^j),$$

can be done component-wise by setting $\lambda_i^j = w_i$ for all j and for all i. But the set of NCAM allocation rules contain more allocation rules than neutral affine maximizers. Further, affine maximizers which does not contain non-zero κ terms are not NCAM allocation rules.

Beyond neutrality. Theorem 1 is no longer true if we drop neutrality. The allocation rule in Section 2.1 is not neutral and cannot be decomposed. Indeed, the set of all implementable allocation rules in this problem includes the set of all almost decomposable allocation rules in Theorem 1 and allocation rules of the form identified in Equation 1 (the affine maximizers of Roberts with non-zero κ terms), and many more, which we have not been able to completely describe.

Payment decomposition. While Theorem 1 establishes that under neutrality an implementable allocation rule can be almost decomposed, can we also decompose payment decisions (almost everywhere)? For a neutral and implementable allocation rule, there will always exists one almost decomposable payment function that implements it - this was shown in Lemma 1. Notice that for every $j \in J$, the payment function defined in Lemma 1, p_i^j depends on the valuation of other components since it is a function of $F^j(\mathbf{v})$. But almost everywhere, F^j can be computed by information of component j (Theorem 1). Hence, almost everywhere, p_i^j only depends on the information of component j. In that sense, the payment function in Lemma 1 is decomposable almost everywhere.

Standard revenue equivalence formula (Chung and Olszewski, 2007) gives a complete description of the set of all payment functions that can implement an NCAM. These are

characterized by a map $h_i : V_{-i} \to \mathbb{R}$ for every agent i. Any payment functions $(q_1, \ldots, q_n)$ that implements an NCAM allocation rule must satisfy for every agent i and every (v_i, v_{-i}),

$$q_i(v_i, v_{-i}) = p_i(v_i, v_{-i}) + h_i(v_{-i}),$$

where p_i is the payment function identified in Lemma 1. Even though, p_i is decomposable (almost everywhere), q_i need not be decomposable since h_i can be chosen in a non-decomposable manner by making it depend on *all* the components of other agents.

Other forms of separability. Our result crucially relies on the fact that valuations are additively separable. The following example shows that other forms of separability may not give us the result. Suppose there are two agents and two components. Further, suppose the valuations satisfy

$$v_i(x) = v_i^1(x^1)v_i^2(x^2), \forall x^1 \in A^1, x^2 \in A^2, i = 1, 2.$$

Clearly, the domain of valuations is separable in the sense that if at a valuation profile v^2 at component 2 agent i prefers alternative x^1 over x'^1, then she continues to prefer the same even when the valuation profile at component 2 changes to v'^2 (because the valuations are positive). However, such valuations are clearly not additively separable.

As is well known, the following affine maximizer allocation rule is implementable (with a fixed order over the alternatives in each component to break ties):

$$F(v) \in \operatorname*{argmax}_{x \in A} \sum_{i=1}^{2} \lambda_i v_i(x) = \operatorname*{argmax}_{x \in A} \sum_{i=1}^{2} \lambda_i v_i^1(x^1) v_i^2(x^2).$$

Let $\lambda_1 = 3, \lambda_2 = 2$. Consider the following two valuations:

	A^1			A^2		
v	a^1	b^1	c^1	a^2	b^2	c^2
v_1	2	3	5	1	2	3
v_2	5	2	1	4	2	1

	A^1			A^2		
$\tilde{v}$	a^1	b^1	c^1	a^2	b^2	c^2
v_1	2	3	5	1	2	3
v_2	5	2	1	4	2	100

The allocation at v is (c^1, c^2), but at $\tilde{v}$, it is (a^1, c^2). Even though, the valuations in component 1 did not change in these two profiles, the outcome changed. This shows that we cannot hope to get the kind of decomposability we got in Theorem 1 beyond additively separable valuations.

4 Proof of Theorem 1

Main idea of the proof. In this section, we prove our main result. Before proceeding to the details of the proof, we give a sketch of the main idea of the proof. In the proof, we fix a component j and fix two profiles v^{-j} and v'^{-j}. Now, F restricted to this component at these two profiles gives two *marginal* allocation rules on component j. These marginal allocation rules are implementable since F is implementable. Because of separable type space structure, the type space restricted to component j satisfies the required richness to invoke earlier results in the literature - in particular, earlier results from the literature imply that both these marginal allocation rules must be neutral affine maximizers. The crux of the proof lies in showing that these two affine maximizers are *equivalent*, i.e., one is obtained by a uniform scaling of the other. This establishes the result. The main idea for establishing this uniform scaling is that if these two affine maximizers are not *equivalent*, then we show that the every mechanism corresponding to F can be manipulated, giving us a contradiction.

Proof: Fix a component j and consider two profiles v^{-j} and v'^{-j}. Denote the restriction of F to profiles of agents where components besides j are fixed at v^{-j} and v'^{-j} as f^j and f'^j respectively. Note that the domain of f^j and f'^j are $V_1^j \times \ldots \times V_n^j$. Since F is implementable, f^j and f'^j are also implementable. Further, the valuation of each agent $i \in N$ for component j lie in $\mathbb{R}_{++}^{|A^j|}$ and $|A^j| \geq 3$. This satisfies all the sufficient conditions for the main result in Mishra and Sen (2012): implementability and neutrality imply that f^j and f'^j are *neutral affine maximizers*, affine maximizers with κ terms set to zero in Equation 1. [8] In other words, there exists $\lambda^j, \lambda'^j \in \mathbb{R}_+^n \setminus \{\mathbf{0}\}$ such that for all $v^j \in V^j$,

$$f^j(v^j) \in \arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i^j v_i^j(a^j).$$
$$f'^j(v^j) \in \arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i'^j v_i^j(a^j).$$

Observation. Notice that it is without loss of generality to assume that λ^j (and λ'^j) is such that for some agent $\ell \in N$, we have $\lambda_\ell^j = 1$. We will use this observation repeatedly at various points in the proof.

[8] Note that the model in Mishra and Sen (2012) is exactly the same as the general model of Roberts (1979) with the set of alternatives denoted as A (which does not have any components). More precisely, let $\mathbb{I}$ be any open interval in $\mathbb{R}$. They show that if type space of each agent is $\mathbb{I}^{|A|}$, then every neutral and implementable allocation rule must be a neutral affine maximizer. This is in contrast to Roberts' theorem, where $\mathbb{I}$ is required to be $\mathbb{R}$. Our result requires additional richness, and we need these intervals to be unbounded from above. Hence, we use $\mathbb{I} = \mathbb{R}_{++}$.

We show that there exists a positive number $\rho > 0$ such that $\lambda_h^j = \rho\lambda_h'^j$ for all $h \in N$. As a consequence of this, the set of maximizers at every v^j is the same for λ^j and λ'^j. Hence, it is without loss of generality to write

$$f'^j(v^j) \in \arg\max_{a^j \in A^j} \sum_{i \in N} \lambda_i^j v_i^j(a^j).$$

This will conclude the proof.

We begin by noting that it is without loss of generality to assume that v^{-j} and v'^{-j} differ in only one agent's valuation - if they differ in more than one agent's valuation, we can repeat this argument to show the desired property. Suppose v^{-j} and v'^{-j} differ in the valuation of agent i, i.e., for all $k \neq j$, we have $v_i^k \neq v_i'^k$ but $v_h^k = v_h'^k$ for all $h \neq i$.

For any alternative a, define a^{-j} as all the alternatives from components other than j-th component. Define Δ as follows:

$$\Delta := \max_{a^{-j}, b^{-j} \in A^{-j}} \sum_{k \neq j} \Big(\big[v_i^k(a^k) - v_i^k(b^k)\big] - \big[v_i'^k(a^k) - v_i'^k(b^k)\big] \Big).$$

Now pick any v^j and v'^j which only differ in valuation of agent i. Suppose $F(v^j, v^{-j}) = a$ and $F(v'^j, v'^{-j}) = b$. Since F is implementable and (v^j, v^{-j}) and (v'^j, v'^{-j}) only differ in the valuation of agent i, we can apply 2-cycle monotonicity across these two valuation profiles - 2-cycle monotonicity is a necessary condition for implementability and can be obtained by adding the pair of incentive constraints of agent i across valuation profiles (v^j, v^{-j}) and (v'^j, v'^{-j}) (Bikhchandani et al., 2006; Vohra, 2011). 2-cycle monotonicity implies that

$$\begin{aligned} &\big[v_i^j(a^j) + \sum_{k \neq j} v_i^k(a^k)\big] - \big[v_i^j(b^j) + \sum_{k \neq j} v_i^k(b^k)\big] \\ &\quad + \big[v_i'^j(b^j) + \sum_{k \neq j} v_i'^k(b^k)\big] - \big[v_i'^j(a^j) + \sum_{k \neq j} v_i'^k(a^k)\big] \geq 0. \end{aligned}$$

Using the definition of Δ, we get

$$\Delta + \big[v_i^j(a^j) - v_i^j(b^j)\big] - \big[v_i'^j(a^j) - v_i'^j(b^j)\big] \geq 0. \tag{2}$$

Now, we complete the step in various cases. Assume for contradiction that there is no $\rho > 0$ such that $\lambda_\ell^j \neq \rho\lambda_\ell'^j$ for all $\ell \in N$. In various cases below, we choose v^j and v'^j appropriately and use Inequality (2) to get a contradiction.

Notation. In the construction of valuation profiles below, we assign zero valuations in many cases. Since 0 is not in the type space (to remind, each $v_i^j \in \mathbb{R}_{++}^{|A^j|}$), what we mean is a valuation arbitrarily close to zero - this is possible since 0 lies in the closure of our type space. We write 0 only to avoid extra notations.

Case 1. For every $h \in N$, $\lambda_h^j \lambda_h'^j = 0$ (i.e, either $\lambda_h^j = 0$ or $\lambda_h'^j = 0$) - note that by definition there is at least one $\ell \in N$ such that $\lambda_\ell^j > 0$ and at least one $\ell' \in N$ such that $\lambda_{\ell'}'^j > 0$. Without loss of generality, we can assume that $\lambda_\ell^j = 1$ and $\lambda_{\ell'}'^j = 1$. Note that (by assumption for this case), $\lambda_\ell'^j = \lambda_{\ell'}^j = 0$. We consider some sub-cases.

Case 1a. Suppose $\lambda_i^j = \lambda_i'^j = 0$. In that case, we construct two valuation profiles of the j-th component - v^j and v'^j such that v^j and v'^j differ from each other by agent i's valuation. Moreover, the valuation of all the agents except agent i and ℓ, ℓ' are zero for all the alternatives in the j-th component, i.e., for all $c^j \in A^j$ we have

$$v_h^j(c^j) = v_h'^j(c^j) = 0 \ \forall \ h \in N \setminus \{i, \ell, \ell'\}.$$

Now, choose $a^j, b^j \in A^j$ and for all $c^j \notin \{a^j, b^j\}$, we have

$$v_i^j(c^j) = v_i'^j(c^j) = v_\ell^j(c^j) = v_\ell'^j(c^j) = v_{\ell'}^j(c^j) = v_{\ell'}'^j(c^j) = 0.$$

Finally, choose $\alpha, \delta, \gamma, \gamma' > 0$ with $\alpha > \delta$ and $K > 0$.

$$v_i^j(a^j) = K(\alpha - \delta), v_i^j(b^j) = v_i'^j(b^j) = 0, v_i'^j(a^j) = K\alpha.$$

$$v_\ell^j(a^j) = v_\ell'^j(a^j) = \gamma, v_\ell^j(b^j) = v_{\ell'}^j(a^j) = v_\ell'^j(b^j) = v_{\ell'}'^j(a^j) = 0, v_{\ell'}^j(b^j) = v_{\ell'}'^j(b^j) = \gamma'.$$

The valuations of agent i, ℓ, ℓ' for alternatives a^j and b^j are shown in Table 1.

	a^j	b^j
v_i^j	$K(\alpha - \delta)$	0
v_ℓ^j	γ	0
$v_{\ell'}^j$	0	γ'
$v_i'^j$	$K\alpha$	0
$v_\ell'^j$	γ	0
$v_{\ell'}'^j$	0	γ'

Table 1: Illustration

By construction, $f^j(v^j) = a^j$ and $f'^j(v'^j) = b^j$. Now, substituting the values of v_i^j and $v_i'^j$ in Inequality (2), we get

$$0 \leq \Delta + K(\alpha - \delta) - K(\alpha) = \Delta - K\delta.$$

Equivalently, we get

$$K\delta \leq \Delta,$$

which is impossible since $K, \delta > 0$ and K can be chosen arbitrarily large. Hence, we get a contradiction for this case.

Case 1b. Suppose $\lambda_i^j > 0$ but $\lambda_i'^j = 0$ (alternately, we can also suppose that $\lambda_i'^j > 0$ but $\lambda_i^j = 0$). As before, there is some $\ell' \in N$ such that $\lambda_{\ell'}'^j > 0$ but $\lambda_{\ell'}^j = 0$. Without loss of generality, we assume that $\lambda_{\ell'}'^j = \lambda_i^j = 1$. In that case, we construct v^j and v'^j very similar to Case 1a with slight modifications.

For all $c^j \in A^j$ we have

$$v_h^j(c^j) = v_h'^j(c^j) = 0 \ \forall \ h \in N \setminus \{i, \ell'\}.$$

Now, choose $a^j, b^j \in A^j$ and for all $c^j \notin \{a^j, b^j\}$, we have

$$v_i^j(c^j) = v_i'^j(c^j) = v_{\ell'}^j(c^j) = v_{\ell'}'^j(c^j) = 0.$$

Finally, choose $\alpha, \delta, \gamma' > 0$ with $\alpha > \delta$ and $K > 0$.

$$v_i^j(a^j) = K(\alpha - \delta), v_i^j(b^j) = v_i'^j(b^j) = 0, v_i'^j(a^j) = K\alpha.$$

$$v_{\ell'}^j(a^j) = v_{\ell'}'^j(a^j) = 0, v_{\ell'}^j(b^j) = v_{\ell'}'^j(b^j) = \gamma'.$$

Valuations of agent i and ℓ' for a^j and b^j are shown in Table 2.

	a^j	b^j
v_i^j	$K(\alpha - \delta)$	0
$v_{\ell'}^j$	0	γ'
$v_i'^j$	$K\alpha$	0
$v_{\ell'}'^j$	0	γ'

Table 2: Illustration

Now, by construction, $f^j(v^j) = a^j$ and $f'^j(v'^j) = b^j$. Substituting the values of v_i^j and $v_i'^j$ in Inequality (2), we get

$$0 \leq \Delta + K(\alpha - \delta) - K(\alpha) = \Delta - K\delta.$$

Equivalently, we get

$$K\delta \leq \Delta,$$

which is impossible since $K, \delta > 0$ and K can be chosen arbitrarily large. Hence, we get a contradiction for this case.

The contradictions in these two sub-cases imply that Case 1 is impossible.

Case 2. There is some agent ℓ such that $\lambda_\ell^j > 0$ and $\lambda_\ell'^j > 0$. Without loss of generality, we assume that $\lambda_\ell^j = \lambda_\ell'^j = 1$. Since there is no $\rho > 0$ such that $\lambda_h^j = \rho\lambda_h'^j$ for all $h \in N$, there is some $\ell' \neq \ell$ such that $\lambda_{\ell'}^j \neq \lambda_{\ell'}'^j$. We now consider sub-cases.

Case 2a. Suppose $\lambda_i^j \neq \lambda_i'^j$. Clearly, either $\lambda_i^j > 0$ or $\lambda_i'^j > 0$. Assume for contradiction $\lambda_i^j > \lambda_i'^j$ - an analogous argument works if $\lambda_i'^j > \lambda_i^j$.

Now, we construct two valuation profiles of the j-th component - v^j and v'^j such that v^j and v'^j differ from each other by agent i's valuation. Moreover, the valuation of all the agents except agent i and ℓ are zero for all the alternatives in the j-th component, i.e., for all $c^j \in A^j$ we have

$$v_h^j(c^j) = v_h'^j(c^j) = 0 \ \forall \ h \in N \setminus \{i, \ell\}.$$

Now, choose $a^j, b^j \in A^j$ and for all $c^j \notin \{a^j, b^j\}$, we have

$$v_i^j(c^j) = v_i'^j(c^j) = v_\ell^j(c^j) = v_\ell'^j(c^j) = 0.$$

Finally, choose $\alpha, \beta, \delta > 0$ with $\alpha > \delta$ and $K > 0$.

$$v_i^j(a^j) = K(\alpha - \delta), v_i^j(b^j) = v_i'^j(b^j) = 0, v_i'^j(a^j) = K\alpha.$$

$$v_\ell^j(a^j) = v_\ell'^j(a^j) = 0, v_\ell^j(b^j) = v_\ell'^j(b^j) = K\beta.$$

The valuations of agent i and ℓ for alternatives a^j and b^j are shown in Table 3.

	a^j	b^j
v_i^j	$K(\alpha - \delta)$	0
v_ℓ^j	0	$K\beta$
$v_i'^j$	$K\alpha$	0
$v_\ell'^j$	0	$K\beta$

Table 3: Illustration

Now, since $\lambda_i^j > 0$, by construction of v^j, if we ensure $\lambda_i^j K(\alpha-\delta) > K\beta$ or $\lambda_i^j(\alpha-\delta) > \beta$, we get $f^j(v^j) = a^j$. Similarly, if we ensure $\lambda_i'^j K\alpha < K\beta$ or $\lambda_i'^j\alpha < \beta$, we get $f'^j(v'^j) = b^j$. Note that by choosing α, β, δ appropriately, we can ensure

$$\lambda_i^j(\alpha - \delta) > \beta > \lambda_i'^j\alpha,$$

since $\lambda_i^j > \lambda_i'^j$. [9] Now, substituting the values of v_i^j and $v_i'^j$ in Inequality (2), we get

$$0 \leq \Delta + K(\alpha - \delta) - K\alpha = \Delta - K\delta.$$

Equivalently, we get

$$K\delta \leq \Delta,$$

which is impossible since $K, \delta > 0$ and K can be chosen to be arbitrarily large. This gives us a contradiction.

Case 2b. Suppose $\lambda_i^j = \lambda_i'^j = 0$. Then, choose ℓ' such that $\lambda_{\ell'}^j \neq \lambda_{\ell'}'^j$. Without loss of generality, assume that $\lambda_{\ell'}^j > \lambda_{\ell'}'^j$.

Now, we construct two valuation profiles of the j-th component - v^j and v'^j such that v^j and v'^j differ from each other by agent i's valuation. Moreover, the valuation of all the agents except agent i, ℓ, ℓ' are zero for all the alternatives in the j-th component, i.e., for all $c^j \in A^j$ we have

$$v_h^j(c^j) = v_h'^j(c^j) = 0 \ \forall \ h \in N \setminus \{i, \ell, \ell'\}.$$

Now, choose $a^j, b^j \in A^j$ and for all $c^j \notin \{a^j, b^j\}$, we have

$$v_i^j(c^j) = v_i'^j(c^j) = v_\ell^j(c^j) = v_\ell'^j(c^j) = v_{\ell'}^j(c^j) = v_{\ell'}'^j(c^j) = 0.$$

Finally, choose $\alpha, \delta, \gamma, \gamma' > 0$ with $\alpha > \delta$ and $K > 0$.

$$v_i^j(a^j) = K(\alpha - \delta), v_i^j(b^j) = v_i'^j(b^j) = 0, v_i'^j(a^j) = K\alpha,$$

$$v_\ell^j(a^j) = v_\ell'^j(a^j) = 0, v_\ell^j(b^j) = v_\ell'^j(b^j) = \gamma,$$
$$v_{\ell'}^j(a^j) = v_{\ell'}'^j(a^j) = \gamma', v_{\ell'}^j(b^j) = v_{\ell'}'^j(b^j) = 0.$$

The valuations of agent i and ℓ for alternatives a^j and b^j are shown in Table 4.

Now, if $\lambda_{\ell'}^j\gamma' > \gamma$, we have $f^j(v^j) = a^j$. Similarly, if $\lambda_\ell'^j\gamma' < \gamma$, we have $f'^j(v'^j) = b^j$. Since $\lambda_{\ell'}^j > \lambda_{\ell'}'^j$, it is possible to choose γ, γ' such that

$$\lambda_{\ell'}^j\gamma' > \gamma > \lambda_\ell'^j\gamma'.$$

[9] A similar argument can be made by switching the roles of v^j and v'^j if $\lambda_i'^j > \lambda_i^j$.

	a^j	b^j
v_i^j	$K(\alpha-\delta)$	0
$v_{\ell'}^j$	γ'	0
v_ℓ^j	0	γ
$v_i'^j$	$K\alpha$	0
$v_{\ell'}'^j$	γ'	0
$v_\ell'^j$	0	γ

Table 4: Illustration

Now, substituting the values of v_i^j and $v_i'^j$ in Inequality (2), we get

$$0 \leq \Delta + K(\alpha-\delta) - K\alpha = \Delta - K\delta.$$

Equivalently, we get

$$K\delta \leq \Delta,$$

which is impossible since $K, \delta > 0$ and K can be chosen to be arbitrarily large. This gives us a contradiction.

Case 2c. Suppose $\lambda_i^j = \lambda_i'^j = \lambda > 0$. Then, choose ℓ' such that $\lambda_{\ell'}^j \neq \lambda_{\ell'}'^j$ - again, such ℓ' exists because there is no $\rho > 0$ such that $\lambda_h^j = \rho\lambda_h'^j$ for all $h \in N$. Without loss of generality, assume that $\lambda_{\ell'}^j < \lambda_{\ell'}'^j$. Then, define

$$\mu := \frac{\lambda}{\lambda_{\ell'}^j}, \mu' := \frac{\lambda}{\lambda_{\ell'}'^j}.$$

Note that $\mu > \mu'$.

Now, we construct two valuation profiles of the j-th component - v^j and v'^j such that v^j and v'^j differ from each other by agent i's valuation. Moreover, the valuation of all the agents except agent i and ℓ' are zero for all the alternatives in the j-th component, i.e., for all $c^j \in A^j$ we have

$$v_h^j(c^j) = v_h'^j(c^j) = 0 \ \forall \ h \in N \setminus \{i, \ell'\}.$$

Now, choose $a^j, b^j \in A^j$ and for all $c^j \notin \{a^j, b^j\}$, we have

$$v_i^j(c^j) = v_i'^j(c^j) = v_{\ell'}^j(c^j) = v_{\ell'}'^j(c^j) = 0.$$

Finally, choose $\alpha, \beta, \delta > 0$ with $\alpha > \delta$ and $K > 0$.

$$v_i^j(a^j) = K(\alpha-\delta), v_i^j(b^j) = v_i'^j(b^j) = 0, v_i'^j(a^j) = K\alpha.$$

	a^j	b^j
v_i^j	$K(\alpha-\delta)$	0
$v_{\ell'}^j$	0	$K\beta$
$v_i'^j$	$K\alpha$	0
$v_{\ell'}'^j$	0	$K\beta$

Table 5: Illustration

$$v_{\ell'}^j(a^j) = v_{\ell'}'^j(a^j) = 0, v_{\ell'}^j(b^j) = v_{\ell'}'^j(b^j) = K\beta.$$

The valuations of agent i and ℓ' for alternatives a^j and b^j are shown in Table 5.

Now, if $\lambda K(\alpha-\delta) > \lambda_{\ell'}^j K\beta$ or, equivalently, $\mu(\alpha-\delta) > \beta$, then we have $f^j(v^j) = a^j$. Similarly, if $\lambda K\alpha < \lambda_{\ell'}'^j K\beta$ or, equivalently, $\mu'\alpha < \beta$, then we have $f'^j(v'^j) = b^j$. Since $\mu > \mu'$, it is possible to satisfy

$$\mu(\alpha-\delta) > \beta > \mu'\alpha,$$

by appropriately choosing α, β, δ with $\alpha > \delta$.

Now, substituting the values of v_i^j and $v_i'^j$ in Inequality (2), we get

$$0 \leq \Delta + K(\alpha-\delta) - K\alpha = \Delta - K\delta.$$

Equivalently, we get

$$K\delta \leq \Delta,$$

which is impossible since $K, \delta > 0$ and K can be chosen to be arbitrarily large. This gives us a contradiction.

This exhausts all the cases and we have thus completed the proof. ■

5 Conclusion

We have investigated an important type space in mechanism design where valuations of agents are additively separable across various components of each alternative. Though implementability does not imply decomposition of allocation decision in each component, by imposing neutrality, we restore decomposition in a generic sense. Our result contributes to the literature on identifying functional forms of mechanisms in quasi-linear environment. More importantly, its implication in separable type space is useful because of decomposability of the allocation rule.

Appendix: Proof of Lemma 1

Proof: Consider (v_i, v_{-i}) and (v_i', v_{-i}). Let F be an NCAM allocation rule satisfying IIA with weights $\{\lambda^j\}_{j\in J}$. Further, suppose $F(v_i, v_{-i}) = a$ and $F(v_i', v_{-i}) = b$. Let $\tilde{J}$ be the set of components such that $\lambda_i^j = 0$ for all $j \in \tilde{J}$. By IIA, $a^j = b^j$ for all $j \in \tilde{J}$. Then,

$$\begin{aligned}
\sum_{j\in J} v_i^j(a^j) + p_i(v_i, v_{-i}) &= \sum_{j\in J} \big[v_i^j(a^j) + p_i^j(v_i, v_{-i})\big] \\
&= \sum_{j\in \tilde{J}} v_i^j(a^j) + \sum_{j\notin \tilde{J}} \big[v_i^j(a^j) + \frac{1}{\lambda_i^j}\sum_{k\neq i} \lambda_k^j v_k^j(a^j)\big] \\
&= \sum_{j\in \tilde{J}} v_i^j(a^j) + \sum_{j\notin \tilde{J}} \frac{1}{\lambda_i^j}\big[\sum_{k\in N} \lambda_k^j v_k^j(a^j)\big] \\
&\geq \sum_{j\in \tilde{J}} v_i^j(b^j) + \sum_{j\notin \tilde{J}} \frac{1}{\lambda_i^j}\big[\sum_{k\in N} \lambda_k^j v_k^j(b^j)\big] \\
&\quad \text{(Using the definition of NCAM and the fact that } a^j = b^j \text{ for all } j \in \tilde{J}) \\
&= \sum_{j\in J} v_i^j(b^j) + \sum_{j\notin \tilde{J}} \big[\frac{1}{\lambda_i^j}\sum_{k\neq i} \lambda_k^j v_k^j(b^j)\big] \\
&= \sum_{j\in J} \big[v_i^j(b^j) + p_i^j(v_i', v_{-i})\big],
\end{aligned}$$

which is the desired inequality for DSIC. ■

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